

PhD Math Camp Exam 2023

Columbia University

August 28, 2023
09:00am

Abstract

This is a timed remote exam. You have time until noon (12pm) to submit your answers via email to the TA (aza2115@columbia.edu) and cc the instructor (mnb2141@columbia.edu). There are eight questions for a total of 120 points (plus one bonus question for an additional 10 points). Please justify your answers, partial credit will be awarded in case of incomplete answers. The exam is closed book and notes, and you are not allowed to consult peers or use the web to aid in solving the problems. A non-programmable calculator is allowed. Columbia's academic honor code applies.

- (15 points) Answer whether each is a distance in $\mathbb{R}^2$ and justify your answer.
 - $d(x, y) := \min \{|x_1 - y_1|, |x_2 - y_2|\}$
 - $d(x, y) := (\|x\| + \|y\|) \cdot \mathbf{1}_{x \neq y}$, where $\|\cdot\|$ is the Euclidean norm (i.e. $\|z\| := \sqrt{z_1^2 + z_2^2}$) and $\mathbf{1}_{x \neq y} = \begin{cases} 1 & \text{if } x \neq y \\ 0 & \text{if } x = y \end{cases}$
- (15 points) Check whether the following sets are subspaces of the n -dimensional real vector space $\langle \mathbb{R}^n, +, \cdot \rangle$, where vector addition and scalar multiplication are defined in the naive component-by-component fashion.
 - When $n = 1$, the set of integers $\mathbb{Z}$.
 - $\text{Ker } A := \{\mathbf{v} \in \mathbb{R}^n : A\mathbf{v} = \mathbf{0}\}$, where A is an $n \times n$ real matrix.
- (15 points) For each of the following functions, show whether the function is homogeneous (if so, state its degree of homogeneity), and state whether it is homothetic and why. If your answer depends on the parameters, explain how it depends on the parameters.
 - $u(x_1, x_2, \dots, x_n) = (\sum_{i=1}^n a_i x_i^\gamma)^{1/\gamma}$.
 - $u(x_1, x_2, \dots, x_n) = a_0 + a_1 x_1 + \dots + a_n x_n$.
- (15 points) Given a normed vector space $(V, +, \cdot, \|\cdot\|)$, show that for any $x \in V$, any open ball centered at x (with respect to the metric induced by the norm) is a convex set.
- (15 points) Let $f : X \rightarrow \mathbb{R}$ be a continuous real function on a metric space (X, d) . Let $Z(f) := \{p \in X : f(p) = 0\}$ be the set of all points in X at which $f(p) = 0$. Prove that $Z(f)$ is closed.

(Hint: You may want to use the sequential characterizations of closed sets and continuous functions as well as the fact that constant sequences converge to the constant itself.)

6. (10 points) Solve the integral

$$\int_2^3 x e^{x+1} dx.$$

Show your steps.

7. (20 points) Consider the following constrained optimization problem:

$$\begin{aligned} \max_{(x,y) \in \mathbb{R}_{++}^2} \quad & \ln(x+1) + \ln(y+1) \\ \text{s.t.} \quad & x + 2y \leq c \\ & x + y \leq 2 \end{aligned}$$

where c is a positive constant.

- Write down the Lagrangian, and the necessary Kuhn-Tucker conditions for a point (x, y) to be a solution of the problem.
 - Solve the problem for $c = \frac{5}{2}$.
 - Let $V(c)$ denote the value function. Find the value of $V'(\frac{5}{2})$ using the Envelope Theorem.
8. (15 points) Consider the correspondence $\Gamma : \mathbb{R} \rightrightarrows \mathbb{R}$

$$\Gamma(x) = \begin{cases} \{0\} & \text{if } x \in (-\infty, -2] \\ [-1, 1] & \text{if } x \in (-2, -1] \\ \{0\} & \text{if } x \in (-1, 0) \\ [0, 1] & \text{if } x = 0 \\ [-1+x, 1-x] & \text{if } x \in (0, 1) \\ \{1\} & \text{if } x \in [1, \infty) \end{cases}$$

- Plot the correspondence.
 - Study the upper and lower hemicontinuity of Γ , i.e. state at which points the correspondence is upper hemicontinuous and at which points it is lower hemicontinuous. (No proof required)
9. **Bonus Question** (10 extra points)

Prove the following statement:

If two finite subsets S and T of the usual inner product space $\mathbb{R}^n$ are strictly separated by a hyperplane (i.e. there exists some $p \in \mathbb{R}^n \setminus \{0\}$ and $c \in \mathbb{R}$ s.t. $S \subset \{x \in \mathbb{R}^n : p \cdot x < c\}$ and $T \subset \{x \in \mathbb{R}^n : p \cdot x > c\}$), then the convex hulls of S and T are disjoint.